Timster's Audio Works - **Split-Flap Clock**

SPLIT-FLAP CLOCK

for Dummies*

The unofficial, slightly cheeky passenger's guide to the desk clock that clacks like a 1970s airport.

For Split-Flap Clock v1.2.0

* Dummies not included. Clatter extremely included. Not affiliated with any famous yellow book series, any airline, any airport, or the Italian company whose departure boards taught a generation of travellers to look up and listen. We just really like them.

DEPARTURES

Or, what's in this book. All gates are a short walk from here.

GATE	DESTINATION	REMARKS
CS	YOUR BOARDING PASS (the cheat sheet)	ON TIME
IC	ANNOUNCEMENTS USED IN THIS BOOK	ON TIME
01	WHAT ON EARTH IS A SPLIT-FLAP DISPLAY?	BOARDING
02	UP AND CLACKING IN UNDER A MINUTE	BOARDING
03	A GUIDED TOUR OF THE BOARD	GO TO GATE
04	DRIVING IT, OR WHERE ALL THE BUTTONS WENT	GO TO GATE
05	THE SETTINGS WINDOW, PAGE BY PAGE	GO TO GATE
06	OTHER PLACES, OR THE ARRIVALS HALL	GATE OPEN
07	REGIONS, OR SORTING YOUR LUGGAGE	GATE OPEN
08	COLOURS, OR THE PAINT SHOP	GATE OPEN
09	DAY AND NIGHT, OR THE VIEW FROM THE WINDOW	GATE OPEN
10	LIGHTING, OR THE LAMPS OVER THE HALL	GATE OPEN
11	REFRESHES, OR THE GREAT CLATTER	FINAL CALL
12	SOUND, OR WHY THE OFFICE HAS GONE QUIET	FINAL CALL
13	ALARMS, OR YOUR WAKE-UP CALL	FINAL CALL
14	LOGIN, DESKTOPS AND OTHER GROUND HANDLING	FINAL CALL
15	ITINERARIES (RECIPES)	GATE CLOSED
16	WHEN THINGS GO WRONG	DELAYED
17	THE PART OF TENS	DEPARTED
GL	PHRASEBOOK FOR THE JET-LAGGED (glossary)	DEPARTED

YOUR BOARDING PASS

Keep this handy. Tear along the dotted line, metaphorically.

HANDS THAT MATTER

Drag the face	Move the clock anywhere
Right-click	The menu: Settings, Keep on top, Hide title bar, Show on all desktops, Sound, Replace the dead bulb (if one's out), Refresh the board, Quit
Double-click	Opens Settings
⚙ top right	Also opens Settings (it appears when the pointer is over the clock)
Resize the window	The board scales to fit and keeps its shape
Alarm ringing	Press the **SNOOZE** or **STOP** cards on the board, or Esc to stop
Click a dark lamp	Puts in a new bulb, when bulbs are replaced by hand
Drag a place	In Settings, Places: moves it on the board, into another region too

WHERE THINGS LIVE

Yellow or white	Settings, **Clock**, Letters
12 or 24 hour	**Clock**, Hours
More places	**Places**, the search box
Groups and order	**Places**: a place's Region, and drag
Wake-up calls	**Alarms**
A whole new look	**Effects**, the moods
The clatter	**Effects**, Motion and Sound
Lamps, bulbs, dimmer	**Effects**, Light
Easier to read	**Effects**, Card brightness
Light or dark	**Clock**, Cards; **Window and About**, Menus
On top, start at login	**Window and About**

PANIC BUTTONS

Too loud	Right-click, untick **Sound**. Then apologise to the open-plan office.
Can't find the window edges	Right-click, untick **Hide title bar**
Clock too big or small	Settings, **Window and About**, Size
Bulbs too dramatic	**Effects**, Bulbs, **Never fail**
Board too dim	**Effects**, the **Dimmer**, or Lamps **Off**
Want the show right now	Right-click, **Refresh the board**
Alarm won't stop	Esc, or right-click, **Stop alarm**

ANNOUNCEMENTS USED IN THIS BOOK

Listen out for these. They are, unlike the real thing, perfectly audible.

PASSENGER INFORMATION Something that makes life easier, or looks better, or both.

PLEASE NOTE Worth writing on the back of your hand. Or your boarding pass.

SECURITY ANNOUNCEMENT Your settings, your colleagues' patience or your sanity are in danger.

CREW ONLY Nerd corner. Skip it freely; the board will keep clacking without you.

GATE 01 WHAT ON EARTH IS A SPLIT-FLAP DISPLAY?

Picture an airport in 1975. Wide lapels, brown carpet, somebody smoking under a sign that says not to, and high above the check-in desks a long black board. Every few minutes the whole thing erupts: hundreds of little cards whirring and clattering at once, letters tumbling, until the noise dies away and there it is, settled and calm. **LONDON 14:35 BOARDING**. Every head in the hall turns up to look.

That was a **split-flap display**. Each character on it is a small drum of hinged cards, a little like a Rolodex. Every card carries the top half of one symbol on its front and the bottom half of the next on its back. Let a card fall and the next symbol appears, top and bottom meeting at the split. To get from A to F, the drum has to flip through B, C, D and E on the way. There's no going backwards. That's why the board spins and why it clatters.

The best known came from an Italian company in Udine, Solari, from the 1950s on, which is why travellers and designers still call them **Solari boards**. The architect Gino Valle, who designed for them, also drew a split-flap desk clock in the 1960s, the Cifra 3, which ended up in New York's Museum of Modern Art. By the 1970s and 80s split-flap boards were in airports and railway stations all over the world. Screens replaced most of them from the 1990s on. Screens are cheaper and easier to update, but they're silent and nobody misses them.

Split-Flap Clock is a love letter to those boards. It doesn't play a video of flaps. Every card is a real little drum in the program: it knows which symbols it carries and in what order, and it can only move forward. Every fall is drawn as it happens, with the top half folding down and the bottom half swinging into place.

> **PLEASE NOTE** Split-Flap Clock is a desk clock, not a timetable. It shows your time, the date, and as many other places as you like, grouped into regions, in the colours you choose. It does not know when your flight leaves. Sorry.

> **CREW ONLY** The minute's tens card has a drum of only six flaps, 0 to 5, just like a real flap clock. So when 10:59 becomes 11:00, both minute cards each fall exactly once: 5 to 0 and 9 to 0. The hour's tens card is 0 to 2 on a 24-hour clock, and blank or 1 on a 12-hour one. Names and dates use the full alphabet drum, which is why they take longest to arrive.

GATE 02 UP AND CLACKING IN UNDER A MINUTE

1. **Collect your ticket.** On the clock's GitHub page, under **Releases**, download **splitflap-clock.exe** for Windows, **splitflap-clock-linux-x86_64** for Linux, or **splitflap-clock-mac.zip** for a Mac.

2. **Check in.** There's nothing to install. On Windows, double-click the exe. On Linux, make the file executable once (`chmod +x splitflap-clock-linux-x86_64`, or tick "Allow executing" in its properties) and then run it. On a Mac, unzip it and drag **Split-Flap Clock** into Applications.
3. **Watch the board come in.** Every card starts blank, then they all set off at once, a moment apart and each at its own speed, until the time and the date settle into place.
4. **Look up.** Feel faintly nostalgic for a decade you may not have lived through. That's normal.

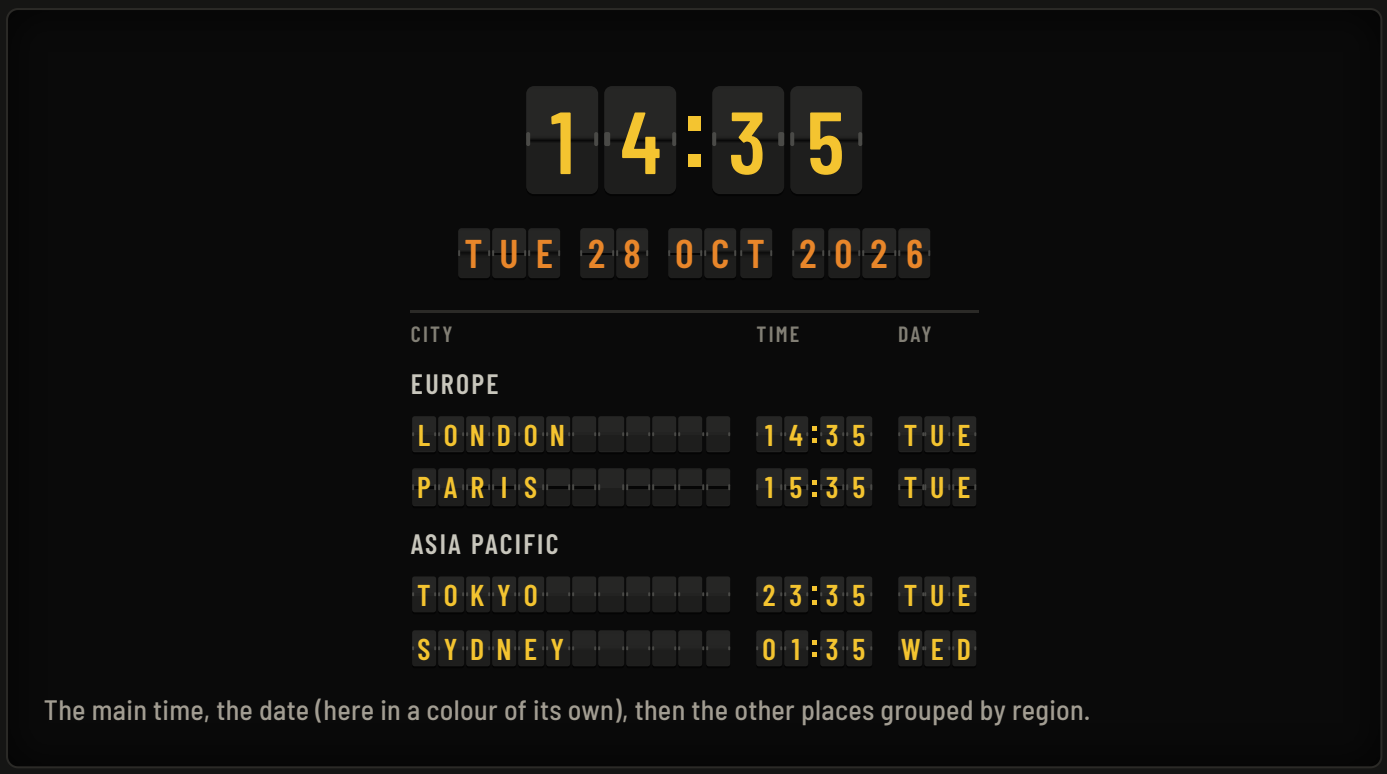
SECURITY ANNOUNCEMENT Windows may say it "protected your PC" from an unrecognised app the first time. The clock isn't signed by a big company, so SmartScreen hasn't heard of it. Click **More info**, then **Run anyway**. It only asks once.

SECURITY ANNOUNCEMENT On a Mac, Apple hasn't signed the clock either, so the first time, right-click it and choose **Open**, then **Open** again. (On newer macOS, try to open it once, then go to System Settings, Privacy & Security, and click **Open Anyway**.) If macOS insists it's damaged, the read-me in the zip has a one-line fix.

PASSENGER INFORMATION The clock is a single file. Put it somewhere you'll keep it, such as a Programs or Apps folder, rather than leaving it in Downloads, especially if you're going to have it start when you log in (Gate 14).

GATE 03 A GUIDED TOUR OF THE BOARD

Here's a board with everything switched on. Yours starts simpler: the time and the date, in yellow.



The main time, the date (here in a colour of its own), then the other places grouped by region.

The main time

Your computer's own time, in big cards: hours and minutes, and seconds if you ask for them. In 12-hour mode the hour's tens card goes blank before 10 o'clock, as on the real thing, and a small **AM** or **PM** card sits beside the minutes.

The date

The day, date, month and year, as in `TUE 28 OCT 2026`. You can switch it off.

Other places

As many as you like, listed under the columns **CITY**, **TIME** and **DAY**. Each has a name of up to 12 cards, its time and its day of the week. The day is how you can tell that it's already tomorrow in Sydney.

The sky

If you switch on **Day and night** (in Effects), a card at the end of each place's row, and one after the main time, shows the sky there: morning, day, evening or night. See Gate 09.

The lamps

If you turn the **Lamps** on (in Effects), a row of warm lamps hangs over the top of the board, and the cards stand out in their light. See Gate 10.

Region headings

If you give places a region, they're grouped under a heading for it, in the order you choose. Places without one go under **OTHER**, unless they come first on the board. See Gate 07.

CREW ONLY The name cards carry the letters A to Z, the digits and a little punctuation: `. - + : / ' & , ( )`. Names are shown in capitals. Anything else, an `é` for instance, still appears, but it arrives in a single flip, because there's no such flap on the drum to come round to.

GATE 04 DRIVING IT, OR WHERE ALL THE BUTTONS WENT

There are no buttons on the board itself. A real board doesn't have any either. Everything is done by hand:

Moving it

Press and drag anywhere on the face. That works with or without a title bar.

Sizing it

Drag the window's edge, and the board scales to fit while keeping its shape. Or use **Size** in Settings (Window and About), which goes from 0.30 to 4.00 times the normal size. When you add places, the window grows to fit them, but never taller than your screen: past that, the whole board shrinks instead.

The menu

Right-click the face for **Settings...**, **Keep on top**, **Hide title bar**, **Show on all desktops**, **Sound**, **Refresh the board** and **Quit**. When an alarm is ringing it has **Snooze 9 minutes** and **Stop alarm** too, and when a bulb is out and waiting for you, **Replace the dead bulb**.

Settings

Double-click the face, choose it from the menu, or click the small ⚙ that appears in the top right corner when the pointer is over the clock. Settings opens in its own window, right beside the clock: to its right, or to its left if the clock is near the edge of the screen. Its pages are down the left side, and along the top is a little copy of the board that shows every change as you make it. Changes take effect straight away; there's no OK button to forget. Gate 05 goes through it page by page.

PASSENGER INFORMATION **Hide title bar** turns the clock into a bare black slab that sits on your desktop like the real thing. You can still drag it about by its face. To make it bigger or smaller without the window edges, use **Size**.

PLEASE NOTE Settings are saved when the clock closes, and every so often while it runs, and loaded next time it starts. They live in `%APPDATA%\splitflap-clock` on Windows, `~/.local/share/splitflap-clock` on Linux, and `~/Library/Application Support/splitflap-clock` on a Mac. Updating to a new version keeps them.

GATE 05

THE SETTINGS WINDOW, PAGE BY PAGE

Settings is laid out like a terminal: the basics in one hall, the show in another. Down the left side are its pages, under three signs. **THE CLOCK** has **Clock**, **Places** and **Alarms**, the things you'll set once and use every day. **THE SHOW** has **Effects**: everything that makes it feel like a real board, none of which changes the time. **THE COMPUTER** has **Window and About**. The number beside Places and Alarms says how many you have.

Along the top of every page is the **preview**, a little copy of the board as your settings make it: the time in its colours, the lamps, the cards' brightness, even the flip speed. On the Places page it shows the place you've opened, and on the Alarms page the alarm you've opened, as it will look when it rings. Each setting has a line under its name saying what it does.

Clock

SETTING	WHAT IT DOES
Hours	**12-hour** or **24-hour**, for every place on the board.
Seconds	Two more big cards on the main time, flipping every second.
Date	The date line under the time, on or off.
Cards	**Black cards**, as on most boards, or **White cards** with dark letters, as on some stations'. See Gate 08.
Letters	**Yellow**, the classic airport amber-yellow, or **White**, the railway-station look, for everything without its own colour. (White cards always have dark letters.)
Main time colour, **Date colour**	Their own colours, or the letters'. See Gate 08.

Places and Alarms

Gate 06 and Gate 07 cover **Places**, and Gate 13 covers **Alarms**.

Effects

At the top are four **moods**, each a whole look in one click, and under them the settings themselves, to fine-tune from there:

MOOD	WHAT IT SETS
Classic	Flat light, a refresh on the hour, and quiet. How the clock comes.
Terminal 1975	Warm lamps, the clatter on, a refresh on the hour, and the cards a touch brighter.

Last flight out	Late evening lamps, slower flaps, refreshes at random, bulbs failing about every hour and replaced by hand, and lighter cards to read them by.
Quiet library	White cards, no sound and no refreshes.

Under the moods it says which one you're in. Change anything and it says **Custom**: your own mix. A mood never touches your places, alarms, colours or the dimmer.

SETTING	WHAT IT DOES
Motion: Flip speed, Refresh	How fast the flaps fall, and how often the whole board does its clatter. See Gate 11.
Sound: Clatter, Volume, Every minute too	The clatter at start-ups and refreshes, how loud, and whether the time ticking over makes a sound too. See Gate 12.
Light: Lamps, Dimmer, Bulbs, Replace by hand, Bulb sounds	The lamps over the board, how bright, how often their bulbs blow, who replaces them, and how loud a bulb is when it goes. See Gate 10.
Cards and sky: Card brightness, Day and night	Lighter cards, so the split and edges show, and the sky card for each place and the main time. See Gates 08 and 09.

Window and About

SETTING	WHAT IT DOES
Keep on top, **Hide title bar**, **Show on all desktops**	How the clock sits among your windows. See Gate 14.
Size	How big the board is, from 0.30 to 4.00 times the normal size.
Menus	The right-click menu and the Settings window: **Like my system**, **Light** or **Dark**.
Start when I log in	See Gate 14.
About	Which version you have and when it came out, and the **Beer money** button (see the Acknowledgements).

> PASSENGER INFORMATION Seconds look wonderful but turn the clock into a busy little machine, flipping every second. For a calm desk, leave them off and let the minutes do the work.

GATE 06 OTHER PLACES, OR THE ARRIVALS HALL

The **Places** page has the search box for adding places, and under it every place on the board, under its region's heading, as the board has them. You can have as many as you like: the board shrinks to fit when they run out of screen.

Adding a place

1. Type into the search box. The clock knows over 7,000 cities and towns, and every time zone, and a list of matches appears as you type, biggest city first:
   - a city: **Manchester**, **São Paulo** (or **sao paulo**; accents are optional);
   - a city and its country, to pick the right one: **Paris, France**;
   - a country, to see its cities: **New Zealand**;
   - or a time zone by its own name, which come after the cities: **Europe/London**.

   Each match says where it is and which time zone it keeps, as in `Sydney, New South Wales, Australia · Australia/Sydney`.
2. Click the one you mean, or don't: the best match is picked already.
3. Press **Add** (it says which, e.g. **Add SYDNEY**) or press Enter. The new place appears at the bottom of the list, opened up so you can change it, and flips onto the board.

> PASSENGER INFORMATION A place added from a city takes the city's name, and its sky card goes by that city's own sunrise and sunset. A place added by its time zone takes the zone's city, which may not be yours: Manchester runs on **Europe/London**. Either way, rename it to whatever you like.

Each place

Each place has a line with its colour, its name, its time zone and the time there now. Click the arrow at the end to open it up:

Name

What the board says, up to 12 characters: **MUM**, **HQ** and **BASE CAMP** are all fine.

Region

Which group it goes in on the board, up to 24 characters. See Gate 07.

Colour

Its own colour, or the letters'. See Gate 08.

Time

From its time zone, the usual, which follows summer time by itself, or **Set by hand**.

Remove

Takes it off the board.

Moving places

Drag a place by the handle at the start of its line, the six dots, and drop it where you want it: a yellow line shows where it will land. Drop it among another region's places and it joins that region. The board rewrites itself from the first row that changed, as a real one would: from where the place went, if you moved it up, or from where it left, if you moved it down. The rows above stay as they are.

Setting a time by hand

Sometimes the time zone isn't the time you want: a ship, a research station, a friend whose clocks run on their own rules, or a place that doesn't match any zone. Choose **Set by hand**. Two boxes appear showing the time there now, hours and minutes. Change them to the time it should be, by typing or dragging, and from then on the clock keeps that place exactly that far ahead of or behind UTC. Next to the boxes it says how far, as in `UTC+05:45, no summer time`.

> PLEASE NOTE A time set by hand stays at that offset all year: it does **not** change for summer time. If the place does, either change it by hand twice a year, or switch back to **From its time zone**.

> CREW ONLY Any clock time can be either side of midnight, as in 23:00 today at UTC+13 or 23:00 yesterday at UTC-11. The clock keeps whichever of the two is nearer to where the place was before you changed it. So nudging the time doesn't flip the day, and the **DAY** column stays right. The list of cities comes from Natural Earth, by way of the city-timezones package; it has the world's cities and big towns, but not every village. For somewhere small, pick the nearest city and rename it.

GATE 07 REGIONS, OR SORTING YOUR LUGGAGE

With a dozen places on the board, it helps to group them. Open a place and type a region into its **Region** box, such as `Europe`, `Clients`, `Family` or `Night shift`, and give the same region to the places that belong together. Or simply drag a place into a region on the Places page. On the board they're gathered under a heading for it.

- Regions are matched without fuss over capitals or stray spaces: `Europe`, `EUROPE` and `europe` are the same region. Headings are shown in capitals.
- Within a region, places keep their order from the Places page.
- Places with no region form a group of their own. If that group is first on the board, it sits straight under the column captions. Anywhere else it gets the heading **OTHER**, so it doesn't look like part of

the region above.

Putting regions in order

On the Places page, each region has a heading with its name and how many places are in it, including **NO REGION** (or **OTHER**). Drag a heading by its handle to move the whole group up or down the board, places and all. Drop a place on a heading to put it at the top of that region.

> PASSENGER INFORMATION Put the region you look at most at the top, just under the date, where your eye lands first.

GATE 08 COLOURS, OR THE PAINT SHOP

Real boards came in one colour, usually yellow or white on black. This one doesn't have to. The main time and the date (on the Clock page), every place and every alarm (when you open them up) can have a colour of their own. Each has three parts:

The colour button
: Click it for the colour picker: drag around the colour square, or type the red, green and blue values into the boxes underneath.

The colour box
: Next to the button, showing the colour as a code like `#F4C430`. Type or paste a colour into it in any of the usual ways: `#F4C430`, `F4C430`, the short `#FC3`, or `244, 196, 48`. The colour changes as soon as what's in the box is a colour; until then it stays as it was.

Reset
: Appears once something has its own colour, and takes it back to the board's yellow or white.

> PASSENGER INFORMATION To give several places exactly the same colour, click the 📋 button in one place's colour picker. It copies the colour as `R, G, B`. Then paste that into the colour box of each of the others. You can also select the code in a colour box and copy it with Ctrl+C.

> PLEASE NOTE Anything without its own colour follows **Letters**. So you can switch the whole board between yellow and white and keep just your favourite places in their own colours.

White cards

Not every board was black. Set **Cards** to **White cards** for light cards with dark letters on a pale grey board, as some railway stations had. Your own colours still apply on white cards, so pick ones that show up on a

light card.

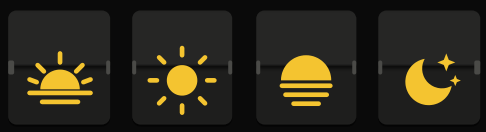
Card brightness

A real board's cards were black on black, which is lovely from across a hall and hard work up close. **Card brightness** (on the Effects page) lifts them: slide it up and the cards turn a lighter charcoal and their hinge pins brighten, so the split across the middle and the edges of each card stand out. On white cards it works the other way round, with a darker split and darker pins. **Classic** takes it back to the usual look.

> SECURITY ANNOUNCEMENT Dark colours on black cards are a fine way to hide a city from yourself. If a place seems to have vanished, check its colour.

GATE 09 DAY AND NIGHT, OR THE VIEW FROM THE WINDOW

A clock tells you the time in Tokyo. It doesn't tell you whether Tokyo is asleep. Switch on **Day and night** on the Effects page, for the **Places**, the **Main time** or both, and each gets one more card: a picture of the sky there now.



Morning, day, evening and night, in the order the card turns.

CARD	WHEN
Morning, the sun rising over the horizon	The hour around sunrise: from half an hour before to half an hour after.
Day, the sun	From then until the hour around sunset.
Evening, the sun sinking into the sea	The hour around sunset: half an hour either side.
Night, the moon and stars	When the sun is down.

The card's flaps go round in the order of a day, so each change is a single flip. On a refresh it spins through all four.

Sunrise and sunset are worked out for each place, for each day, from where it is: the city you found it by, or its time zone's city. That's built into the clock, so nothing is looked up online. So the night really is most of

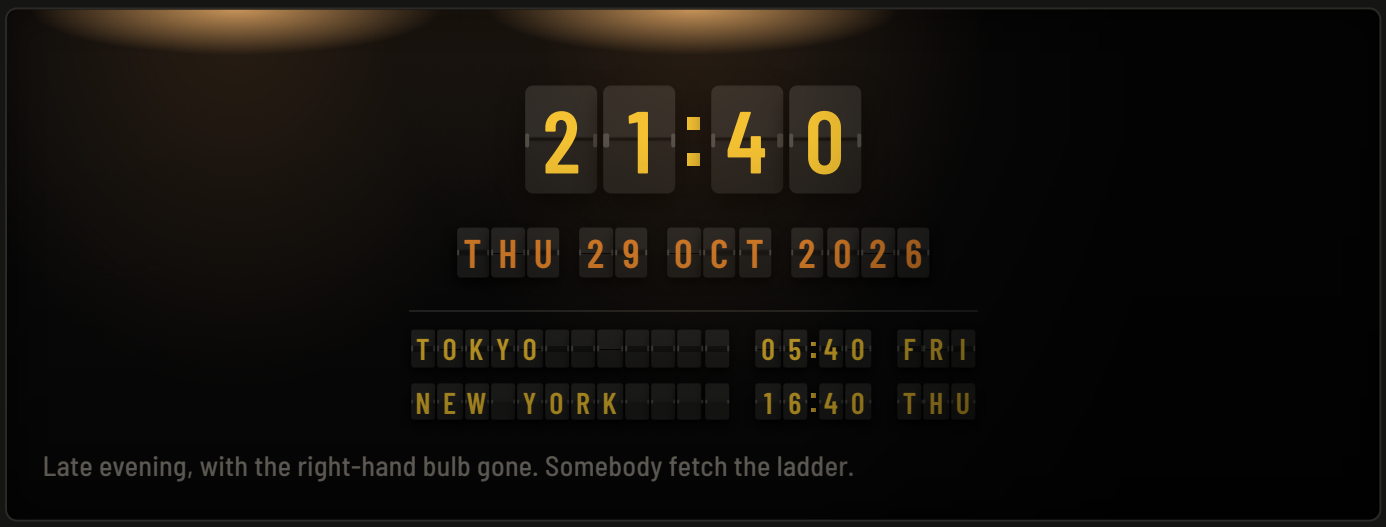
the day in Reykjavik in December, and summer at Svalbard never gets dark at all. The main time's card uses the city of your computer's own time zone.

> **PLEASE NOTE** A place whose time is **set by hand**, or a zone with no city, such as UTC, has no location to go by, so its sky goes by its clock: sunrise at six in the morning and sunset at six in the evening.

> **CREW ONLY** The sums are the usual ones for the sun's position, good to a minute or so. For a place added by its time zone, the zone's city stands for the whole zone, so for somewhere far from it, the card can change a little before or after the sun really does. Add it by its city instead, and it's right.

GATE 10 LIGHTING, OR THE LAMPS OVER THE HALL

Departure halls were never evenly lit. The boards hung under rows of warm incandescent lamps, each throwing its own pool of light, so the cards near a lamp glowed and the ones between them sat in shadow. And every so often a bulb would give up, and a stretch of the board went dark until somebody came with a ladder. Turn the **Lamps** on, on the Effects page, and the clock does the same.



Late evening, with the right-hand bulb gone. Somebody fetch the ladder.

LAMPS	THE LOOK
Off	No lamps: the board evenly lit, as it always was. This is how the clock comes.
Soft	A gentle warm wash, the lamps only just showing.
Warm	A clear pool of light under each lamp, fading down the board.
Evening	The hall lights down, the lamps doing all the work: dimmer, more amber, with deep shadows.

The lamps hang just over the top edge, one for every few cards' width, so a wide board gets more of them. Each card throws a soft shadow down onto the board behind it, deeper the dimmer the lamps. It's what

makes the board look like a thing hanging on a wall rather than a picture of one.

The dimmer

Dimmer turns the lamps down to a quarter, for a hall at closing time, or up to 160%, for a hall at noon. The light of the hall itself stays as it is; only the lamps change. **Normal** puts it back. The moods leave the dimmer alone, so once you've found your level, it stays.

Bulbs

Choose how often bulbs fail: **Fail every few days** (the usual), **Fail about once a day**, **Fail about every hour**, for the impatient, or **Never fail**. When is left to chance, so a bulb might go minutes after the last one, or not for a week. Only the time the clock is running counts. Can't wait at all? **Blow one now** sets one off.

A failing bulb flickers and stutters for a few seconds. Most of the time it steadies and carries on. Sometimes it blows:

1. A white-hot flash, and a pop, with a dull thump behind it.
2. Its glass flies out and tumbles to the foot of the board, landing in a little brittle scatter of ticks a moment after the pop, and lies there in a small heap, the odd piece catching the light of the other lamps.
3. A thin trail of smoke curls out under the top of the board, and the draught carries it off sideways until it's gone, in a few seconds.
4. Its part of the board goes dark.

Bulb sounds sets how loud the pop and the glass are, on their own: they play whether the clatter is on or not, and **Off** silences them.

Replacing them

Replaced for you
: The usual, with **Replace by hand** off. Someone comes round with a new bulb after a minute or so, and the new one sputters on.

Replace by hand
: A dead bulb stays out, and its glass stays where it fell, until you do something about it, even if you close the clock and start it again. Move the pointer over the dark lamp at the top of the board: it turns into a hand and says **Click to put in a new bulb**. Click, and the new one sputters and comes on. Or right-click the clock and choose **Replace the dead bulb** (or **Replace the 2 dead bulbs**, if you've let things slide).

The janitor

A new bulb is only half the job. Once it's in, the janitor comes along the foot of the board with a broom, from behind the heap of glass, and sweeps it off the nearer end of the board, a stroke at a time, the pieces skittering ahead of the bristles. Then he's gone, and so is the glass. Good help is hard to find; this one never misses a shard.

PASSENGER INFORMATION **Evening** lamps with **Card brightness** turned up is the easiest dim board to read: the lamps set the mood and the lighter cards keep the letters crisp. The **Last flight out** mood does just that.

PLEASE NOTE The lighting is only a look. A dark patch never hides the time: the cards under a dead bulb still flip, just in shadow. If the whole board looks dim, turn the **Dimmer** up, or the lamps **Off**.

CREW ONLY The bulbs behave like filaments: they take a few hundredths of a second to warm up and cool down, which is why a flicker looks like a stutter rather than a switch. The light is drawn over the finished board, so it falls on the cards, their letters and the board alike. The pop and the glass are made as they play, like the clatter, and are heard from the side of the board the bulb was on.

GATE 11 REFRESHES, OR THE GREAT CLATTER

Every so often, a real board would refresh: every card spinning round its drum at once, and then everything landing back where it was. Split-Flap Clock does the same. On a refresh, every card makes one full turn of its drum, through every symbol it carries, and lands back on the right one.

Like a real board, it's not tidy. The cards all start a moment apart and each runs at its own speed, so the board scrambles all over and settles card by card. The same happens when the clock starts.

REFRESH (EFFECTS, MOTION)	WHEN
Never	Only when you ask (see below).
Every 15 minutes	On the quarter-hours by your clock: :00, :15, :30, :45.
Every 30 minutes	On the hour and the half-hour.
Every hour	On the hour. This is how the clock comes.
At random	After a random wait, between two times you choose: **Between** 10 **and** 60 **minutes apart** to start with, anything from 1 to 720. Each wait is drawn afresh, so you never know quite when it's coming.

A place on the move

Move a place on the Places page and the board refreshes too, but only from the first row that changed down, the way a real board rewrites only what's moved. See Gate 06.

Flip speed

Flip speed, on the Effects page, sets how fast the flaps fall, from a quarter as fast as usual, for a stately board, to four times as fast, for a frantic one. It sets everything that moves: each flap's fall, the spin through the alphabet, and the ripple across the board. **Normal** takes it back to the usual speed.

> PASSENGER INFORMATION Want one now? Right-click and choose **Refresh the board**. It's the most satisfying thing in this book.

> CREW ONLY The quick flips on the way take about 0.07 seconds each, and the last one, landing on the right symbol, takes 0.3 seconds. A name card has 47 flaps on its drum, so a full turn takes it three seconds or so. The digit cards, with only ten flaps or fewer, are done in a moment. That's why the names are always last to settle, on a real board too.

GATE 12 SOUND, OR WHY THE OFFICE HAS GONE QUIET

A split-flap board without the noise is only half the experience. Switch on **Clatter** on the Effects page (or right-click and tick **Sound**) and every flap that falls makes its click. There are no recordings: each click is made as it plays, so no two are quite the same.

- Big cards sound deeper and heavier than small ones.
- Each card is heard from its place on the board, left to right, if you have stereo speakers.
- The last flap of each card lands with a slightly firmer knock.
- A refresh builds to a proper clatter, then thins out to the last few stragglers.

Volume sets how loud. The clock itself is quiet: only the start-up and refreshes make a sound, never the minutes ticking over, unless you also switch on **Every minute too**.

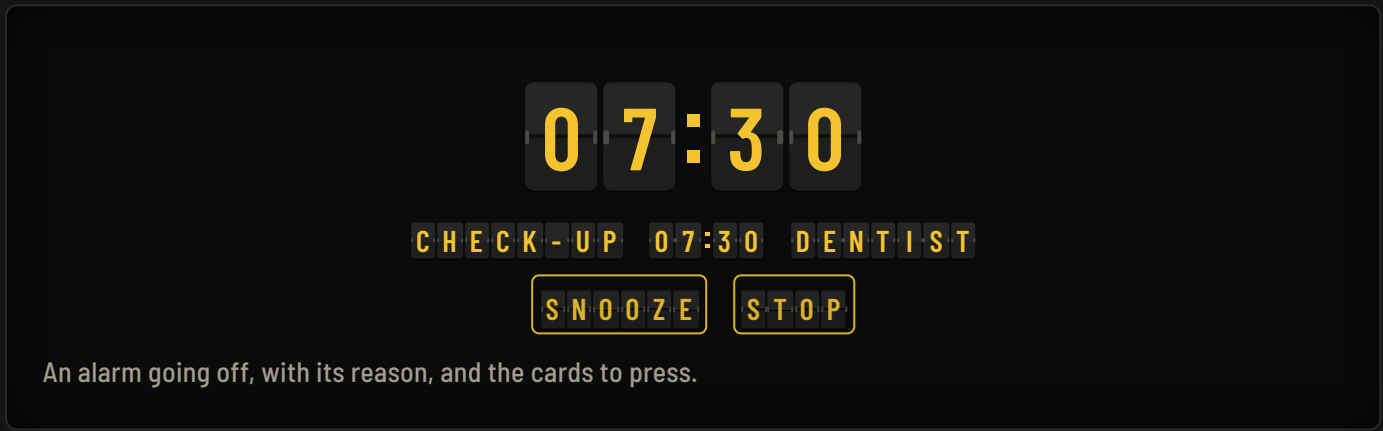
Two other sounds have volumes of their own, and play whether the clatter is on or not: the alarm chime (**Chime volume**, on the Alarms page) and a bulb blowing (**Bulb sounds**, in Effects, Light).

> SECURITY ANNOUNCEMENT **Every minute too**, with **Seconds** on, makes a sound every single second. Some people find that soothing. Their colleagues may not.

> **PLEASE NOTE** The clatter is off when the clock comes. If there's no sound output to play on, Settings says so in orange, and the clock carries on silently. It keeps trying every 15 seconds, so if your speakers come back, or the sound system wakes up after you log in, the clatter returns by itself.

GATE 13 ALARMS, OR YOUR WAKE-UP CALL

An airport never lets a flight leave without telling you. Neither does Split-Flap Clock. Set an alarm and, when it's time, the board announces it: a clatter, a row of its own flipping in under the time, and a proper airport chime, *bing-bong*, every few seconds until you answer it.



An alarm going off, with its reason, and the cards to press.

Setting one

On the **Alarms** page, press **Add an alarm**. Each alarm has a line: a switch to turn it on or off, its time in big figures, its name and reason, when it rings and by whose clock, the days it rings on, lit, and its colour. Click the arrow at the end to open it up:

Time

The hour and the minute, which you can type into or drag.

Name

The alarm's name on the board when it rings, up to 12 characters: **WAKE UP**, **TEA**, **CALL MUM**.

Reason

What it's for, such as **DENTIST** or **PICK UP KIDS**, also up to 12 characters. When it rings, the end of its row says the reason. With no reason it says **NOW**.

Colour

Its own colour, so you can tell the dentist from the school run at a glance. When it rings, its row and its **SNOOZE** and **STOP** cards are in that colour. **Reset** goes back to the board's. See Gate 08 for how the colour box works.

Repeat

Click the days it rings on: **M T W T F S S**. The days you've picked light up yellow, and it says what you've picked, such as **Weekdays**, **Weekends** or **Every day**. With no days picked it says **Once**: the alarm rings the next time its time comes round, then turns itself off.

On a date

Switch it on to ring once on a particular day instead, such as an appointment next month: set the day, the month and the year. It says **Once, on** and the date. Afterwards it turns itself off.

Time at

Your time, or any place on the board. An alarm at 09:00 at TOKYO rings when it's nine in the morning in Tokyo, summer time and all. It keeps following that place wherever you move it. If you remove the place, the alarm goes back to your time.

Test

Rings it straight away, so you can hear what you're in for.

Remove

Gets rid of it.

The preview along the top shows the alarm you've opened as it will look on the board when it rings. To remove an alarm without opening it, click the little cross on its line: it asks **Remove this alarm?**, with **Remove** and **Keep it**, so nothing goes by accident.

Under the alarms, **Chime volume** sets how loud the chime is. The chime has its own volume, and it rings even if the clatter is off, because an alarm you can't hear isn't much use.

When it goes off

- The board does a refresh, with its clatter if the clatter is on, and a row flips in under the time with the alarm's label, its time and its reason, or **NOW**. Under it come two cards to press, **SNOOZE** and **STOP**, which light up under the pointer.
- The chime plays every four seconds.
- **SNOOZE** (or a click anywhere else on the board) makes it go quiet for nine minutes. The row says **DELAYED** with the time it will ring again. Then it all starts again.
- **STOP** stops it. So does Esc, or right-clicking and choosing **Stop alarm**. The right-click menu has **Snooze 9 minutes** too.

> PLEASE NOTE The clock has to be running for its alarms to ring. It can't wake a computer that's asleep or switched off. An alarm only rings for times that come round while the clock is running. Starting the clock just after an alarm's time doesn't ring it, and neither does setting or changing an alarm for a time that's just gone. If the computer was asleep when an alarm was due, it still rings when it wakes, as long as that's within 10 minutes. And an alarm that nobody answers gives up by itself after half an hour, so you won't come back from holiday to a clock that's been chiming for a fortnight.

PASSENGER INFORMATION Want to hear an alarm now? Press **Test**. Setting one for the minute it already is won't ring until that time comes round again. Changed an alarm's time after it already rang today? It can ring again today, at its new time.

CREW ONLY Why nine minutes? Because that's what every snooze button has done since the 1950s, and who are we to argue. The chime is two bell notes, E then C, made from a few pure tones as it plays, like the clatter. If two alarms go off together, the board shows the first, and a snooze or a stop answers both.

GATE 14 LOGIN, DESKTOPS AND OTHER GROUND HANDLING

Start when I log in

Switch it on in Settings (Window and About) and the clock starts with you every time you log in. On Windows it appears under Startup apps in Task Manager, where you can also switch it off. On Linux it adds an entry to `~/.config/autostart`, and on a Mac a login item in `~/Library/LaunchAgents`. If you move the clock's file to another folder, the entry is updated the next time you run it from there.

Show on all desktops

If you use more than one virtual desktop or workspace, this keeps the clock on every one of them, so it's there wherever you switch to.

- **Windows 10 and 11:** the clock has no taskbar button while this is on, and doesn't appear in Alt+Tab. That's the price Windows asks for being on every desktop. Right-click the clock to get at it.
- **Linux:** it asks the desktop to keep the clock on every workspace, in the standard way that desktops with workspaces understand.
- **Mac:** the Dock decides. Right-click the clock in the Dock, then **Options**, **Assign To**, **All Desktops**.

Keep on top

Keeps the clock above other windows, like a board above a busy hall. These three, **Keep on top**, **Hide title bar** and **Show on all desktops**, are on the Window and About page, and in the right-click menu too.

CREW ONLY On Linux the clock runs as an X11 window wherever it can, through XWayland on Wayland desktops. Wayland doesn't let apps keep themselves on top or on every workspace, but X11 windows can. The only side effect is that with fractional display scaling, such as 125%, the clock can look a touch soft.

GATE 15 ITINERARIES (RECIPES)

TERMINAL 1, 1975

The **Terminal 1975** mood, Letters **Yellow**, **24-hour**, Date on, the clatter at a modest volume. Add three far-off places and give them all the region **Departures**. Make a cup of tea at five to the hour and come back for the show.

THE RAILWAY CONCOURSE

Letters **White**, **24-hour**, Seconds on, Refresh **Every 15 minutes**. Hide the title bar and stretch the clock long and low across the top of your screen with **Size**. Mind the gap.

THE WORLD DESK

Add every office you deal with. Give them regions such as **Americas**, **Europe** and **Asia Pacific**, and drag your own region's heading to the top. Colour the ones that are awake right now in green, and enjoy feeling like mission control.

FAMILY ABROAD

Search for each family member's town and rename it to the person: **MUM**, **JAMES**, **GRANDAD**. Give them the region **Family** and a colour each. You'll never ring anyone at 3 a.m. again. Probably.

THE SURPRISE

Refresh **At random**, **Between** 5 **and** 25 **minutes apart**, the clatter on. Show on all desktops. Every so often the clock clatters into life all by itself, and everybody looks up, just like they used to.

THE EARLY FLIGHT

An alarm at 06:45 on **Weekdays**, labelled WAKE UP, with the chime volume up. Another at 08:30, labelled LEAVE NOW. The board pages you like a passenger who's been in the duty-free too long.

THE CALL TO TOKYO

Add Tokyo as a place, then an alarm at 09:00, **Time at** TOKYO, labelled CALL TOKYO, on Monday to Friday. It rings when Tokyo's day starts, whatever your clocks are doing for summer time.

THE LAST FLIGHT OUT

The **Last flight out** mood, the **Dimmer** down to about 70%, **Bulb sounds** up. Dim the room. When a lamp goes, leave it and its glass a while: the hall feels emptier for it.

THE QUIET LIBRARY

The **Quiet library** mood, Seconds off, **12-hour**, **Bulb sounds** off. The cards still flip, silently, once a minute. Very civilised.

GATE 16 WHEN THINGS GO WRONG

Windows won't open it ("Windows protected your PC")

Click **More info**, then **Run anyway**. The clock isn't signed, so SmartScreen doesn't recognise it.

Mac: it won't open, or says it's damaged

Right-click it and choose **Open**, or use **Open Anyway** in System Settings, Privacy & Security. If it says it's damaged, open Terminal and run `xattr -cr "/Applications/Split-Flap Clock.app"`, then open it again.

Mac: no ⚙ in the corner

On a Mac, use the right-click menu or a double-click for Settings instead.

Mac: Show on all desktops does nothing

On a Mac, that's the Dock's job: right-click the clock in the Dock, then **Options**, **Assign To**, **All Desktops**.

Linux: nothing happens when I open it

Make the file executable first: `chmod +x splitflap-clock-linux-x86_64`. It also needs a desktop with OpenGL, and ALSA for sound, which every ordinary desktop has, PipeWire and PulseAudio ones included.

The time is wrong

The main time is your computer's own clock and time zone. Fix those in your system's date and time settings, and the board follows.

One place is an hour out

Check whether it's **Set by hand**. Hand-set times don't change for summer time. Switch to **From its time zone**, or check you picked the right zone.

No sound

Is **Clatter** on, on the Effects page, and the **Volume** up? The clock itself is silent apart from start-ups and refreshes, unless you switch on **Every minute too**. If Settings shows an orange message, the clock can't find a sound output. It tries again every 15 seconds.

The window has no edges and I can't resize it

That's **Hide title bar**. Use **Size** in Settings (Window and About), or right-click and untick **Hide title bar**.

I can't find the clock in the taskbar

On Windows, **Show on all desktops** removes its taskbar button. It's still there: look on the desktop and right-click it.

"Show on all desktops" says Wayland doesn't let apps choose this

The clock is running as a native Wayland window, which happens when your session has no X11 (XWayland) at all. Use your desktop's own window menu, often a right-click on the title bar, to put it on all workspaces.

An alarm didn't ring

Is it ticked on, with the right days lit? A **Once** alarm, or one **On a date**, turns itself off after it has rung. Was the clock running? It can't ring while the computer is asleep or the clock is closed, and it only makes up for a missed alarm within 10 minutes of the computer waking. It doesn't ring for a time that had already passed when the clock started or when you set it. Is **Time at** set to the place you meant?

A sky card shows the wrong time of day

Is the place set by hand? Then it goes by its clock, with sunrise and sunset at six. Otherwise it goes by the sun at its time zone's city, which for somewhere far from that city can be a little early or late.

The alarm's row is small

The row fits within the board's width, so the window doesn't jump about when it appears. On a board with only the time on it, that means small cards. Add a place or the date, and it gets bigger.

Part of the board has gone dark

A bulb has died (Gate 10). With **Replace by hand** it stays out until you click the dark lamp at the top, or right-click and choose **Replace the dead bulb**. Bulbs don't fail at all with **Never fail**, or with the lamps off.

The whole board looks darker (or brighter) than it was

Did you pick a mood? **Terminal 1975** and **Last flight out** change the lamps and the cards' brightness in one go. The **Dimmer** sets how bright the lamps are, whatever the mood.

I heard a pop with the clatter off

That was a bulb. Bulbs have their own volume, **Bulb sounds** in Effects, Light; slide it to **Off** for silence.

The search can't find my town

It knows the world's cities and big towns, not every village. Search for the nearest city, add it, and rename it. Or search for the time zone, such as `Europe/Berlin`.

The cards are hard to make out

Turn up **Card brightness**. If the lamps are on, turn the **Dimmer** up, or try **Soft** or **Off**.

A place has disappeared

Check its colour: dark grey on dark cards is very discreet. Or it may be further down than you think, under its region's heading.

My settings didn't stick

They're saved when the clock closes normally. If the computer crashed or the clock was killed, it may have missed the last few changes.

GATE 17 THE PART OF TENS

Ten things that are not delays (honest)

1. Names take longer to settle than digits. Their drums have 47 flaps; the digits' have ten at most.
2. A card only ever goes forward. To go from B to A it goes the long way round, through all of C to Z, the digits and the punctuation.
3. When the minute goes from 59 to 00, both cards flip exactly once. The tens drum only has 0 to 5.
4. The hour's tens card is blank before 10 o'clock on a 12-hour clock. Real flap clocks did that.
5. An `é` arrives in a single flip. It isn't on the drum, so it has nothing to come round from.

6. A dark patch on the board is a dead bulb, not a dead clock. The cards under it are still flipping, in the shadows.
7. The start-up is never quite the same twice. The cards set off a moment apart, at their own speeds, on purpose.
8. The clock makes no sound on the minute unless you ask it to. It's a desk clock, not a metronome. The snooze is nine minutes, because every snooze is nine minutes.
9. On Windows, no taskbar button with **Show on all desktops**. That's how Windows does it.
10. The board gets smaller as you add places. The window stops growing at the screen's edge, so the board scales down to fit.

Ten ways to annoy your office with Split-Flap Clock

1. Refresh **At random**, between 1 and 2 minutes apart, with the sound up.
2. Rename everyone's home city to **DELAYED**.
3. Add a place called **LUNCH**, set by hand two hours ahead, and insist it's lunchtime there.
4. An alarm called **BOARDING** at 16:55 every weekday, chime volume all the way up.
5. Give every place a slightly different shade of beige. Call it "1970s accuracy".
6. Bulbs **Fail about every hour**, **Replace by hand**, bulb sounds all the way up, and tell everyone the glass is facilities' job.
7. Make a region called **GATE CLOSED** and put your manager's city in it.
8. Press **Blow one now** just as your manager walks past, and look up at the ceiling.
9. Add **MARS**, set by hand, and refuse to explain.
10. Choose **Refresh the board** whenever anyone asks you a question, and wait for it to finish before answering.

PHRASEBOOK FOR THE JET-LAGGED

Split-flap display

A display made of drums of hinged cards that fall one by one to show the next symbol. Also called a flap board or a Solari board.

Card, or module

One character's drum of flaps. The clock's main time has four of them, or six with seconds.

Drum

The set of symbols a card carries, in order. Digits have 10, the minute's tens 6, names and dates 47.

Flap

One hinged leaf, carrying the top half of one symbol and the bottom half of the next.

Sky card

The card showing morning, day, evening or night at a place, from its sunrise and sunset.

Lamp

One of the warm lights over the board, when the lamps are on. Its bulb can flicker, or blow.

Janitor

Never seen, but his broom is: it sweeps a blown bulb's glass away once the bulb is replaced.

Mood

A whole look in one click, on the Effects page: lamps, sound, refreshes, flip speed and cards together.

Dimmer

How bright the lamps are, from a quarter to 160%.

Preview

The little board along the top of Settings, showing your changes as you make them.

Snooze

Nine more minutes. Click the board while an alarm rings.

Refresh

Every card spinning once round its drum and landing back where it was.

Region

Your name for a group of places on the board, with a heading of its own.

Time zone

An area that keeps the same clock, named after a big city in it, such as `Europe/Paris`. It knows its own summer time rules.

UTC

Coordinated Universal Time, the world's reference clock, near enough Greenwich Mean Time. Every other time is so many hours and minutes ahead of it or behind it.

Offset

How far a place is ahead of UTC or behind it, as in UTC+05:30. A time set by hand is a fixed offset.

Summer time (DST)

The hour many places move their clocks forward in summer, and back again in autumn. Time zones follow it; hand-set times don't.

Virtual desktop, or workspace

Separate screens' worth of windows on one computer, switched between with a keypress or gesture.

Hex code

A colour written as `#RRGGBB`: red, green and blue, each from 00 to FF.

XWayland

The part of a Wayland desktop that runs older X11 programs. The clock uses it on Linux so that it can stay on top and on every workspace.

ACKNOWLEDGEMENTS

Special thanks to **Oscar**, our most particular passenger. He asked for unlimited places, times set by hand, colours for everything, and regions to sort them into. Then he tested the lot and spotted that you could copy a colour but not paste one. Then he asked for the sky, the reasons, the dates, the white cards, the buttons, the flip speed and the card brightness. And then for bulbs that blow, with glass and smoke, a dimmer to see them by, and a janitor to sweep up after them. Much of version 1 is his doing, and nearly all of 1.1 and 1.2. Every board needs someone who reads it properly.

Thanks too to **Craig** and **Nicholas**, who took the first Mac version for a spin on the very latest macOS and reported back: sound, keeping on top, the bare slab, every desktop and the Settings window all present and correct. The Mac version is here because of them.

And thanks to **my Dad**, who tried to add a place, couldn't find it, and said so: the clock needed a much better list of places than the world's time zones. Thanks to him it knows over 7,000 cities and towns, and finds Manchester, São Paulo and the whole of New Zealand by name.

Beer money. If you like Split-Flap Clock and want to support its developer with some beer funding, send it via PayPal to spoolbeermoney@outlook.com. There's a **Beer money** button in Settings too, under Window and About. Cheers!



This concludes our tour of the board. Please make sure you have all your belongings with you.

Thank you for flying Split-Flap Clock.